

VibeGuard: Amith's Campus Execution & Faculty Requisition Action Guide

Assigned To: Amith Krishna Das (Hardware & Fabrication Lead)

Project: VibeGuard — Edge AI Industrial Vibration Monitoring & Fault Diagnosis

Institution: Jyothi Engineering College (Autonomous), Cheruthuruthy, Kerala

Target Execution Date: Today / Immediate Campus Turnaround

Available Cash Reserve: Supported from project fund (Est. total out-of-pocket: ~₹150)

1. Quick Mission Briefing

Amith, you have already received the main electronic hardware package (ESP32, motor, sensor, power adapter, and jumpers). Your mission today is to visit **three campus facilities** and pick up **four small hardware items** so our physical test rig can be fully assembled:

[STOP 1: FabLab] ET Store]	[STOP 2: Mech Workshop]	[STOP 3: ECE Lab /
- 3D Print Rotor Arm (x2)	- Cut 15x15 cm Wood Baseplate	- MB102 Breadboard
- 3D Print Sensor Mount (x1)	- Drill 2mm pilot holes	- M3 Bolt,
Washers, Nyloc Nut		
Cost: ~₹50.00	Cost: ₹0.00 (Scrap Bin)	Cost: ₹0.00 (Lab)
or ~₹105 (ET Store)		

2. Stop 1: College FabLab (3D Printing & Prototyping)

- **Location:** FabLab / Makerspace (near Innovation & Incubation Center).
- **Whom to Meet:** FabLab Student Coordinator, 3D Printing Operator, or Lab Technician.
- **Exact Files to Give Them (on USB Pen Drive or WhatsApp):**

1. [VibeGuard Rotor Arm D Shaft.stl](#) — *The rotating cam arm 3D model.*
2. [VibeGuard ADXL345 Rigid Mount.stl](#) — *The sensor bracket 3D model.*
3. [VibeGuard College FabLab 3D Printing Specification.pdf](#) — *The official spec sheet with dimensions, infill rules, and billing slip.*

What to Say to the FabLab Operator (Dialogue Script):

"Good morning / Chetta, we are working on our semester microcontroller project, VibeGuard. We have two small 3D parts ready in STL format for our vibration test rig:

1. **Rotor Cam Arm (print 2 copies):** It needs **80% to 100% infill** in PLA because it spins on a motor shaft at 600 RPM.

2. **ADXL345 Sensor Bracket (print 1 copy):** Standard **60% infill** in PLA.

Total weight is only around **12 to 13 grams**, so at the college rate of ₹4/gram, the bill is around **₹50.00**. Here are the STL files and the specification sheet with the billing slip."

Key Technical Things to Tell Them:

- **Material:** Standard PLA (Black, Blue, or Grey).
- **Perimeters / Walls:** Minimum **4 wall loops** so the 3.0 mm D-shaft bore is solid plastic.
- **Estimated Print Time:** ~35 to 40 minutes total.

Acceptance Checklist Before Leaving FabLab:

- ☐ Take your N20 motor shaft from your pocket and push it into the D-bore hole of the printed arm. It should slide on firmly with **no loose wobble**.
 - ☐ Ensure the 3.2 mm hole on the outer end is clean and free of plastic burrs.
 - ☐ Collect both rotor arms (1 primary + 1 spare) and the sensor bracket.
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3. Stop 2: Mechanical Workshop & Carpentry Section

- **Location:** Mechanical Engineering Workshop (Carpentry & Fitting Hall).

- **Whom to Meet:** Carpentry Instructor / Workshop Superintendent.
- **Exact File to Show / Give (Show on Phone or Print PDF):**

* [VibeGuard College Mech Workshop Baseplate Requisition.pdf](#)

What to Say to the Carpentry Instructor (Dialogue Script):

"Good morning Sir,

We are building a machine vibration monitoring testbed for our semester project. We need a small, rigid baseboard from the workshop scrap bin:

*1. Could you help us cut an offcut piece of **plywood or dense hardwood, roughly 15 cm by 15 cm (6 inches square), 10 to 12 mm thick?***

*2. We also have a miniature metal motor bracket. Could you help us drill two small **2.0 mm pilot holes spaced 17 mm apart** so our screws don't split the wood?*

*3. Do you have **2 to 4 small self-tapping wood screws (~10 mm length)** to hold the bracket into the wood?*

Here is the layout diagram on my phone, Sir."

Key Technical Rule:

- **Why thick plywood?** The baseplate must be **at least 10 mm thick**. If they offer thin 3 mm ply or thin sheet metal, politely ask for thicker plywood offcut. Thin sheets vibrate like a drumhead and ruin sensor data.

Acceptance Checklist Before Leaving Workshop:

- ☐ The wooden plank rests flat on a table with **zero rocking or wobble**.
 - ☐ Place your metal N20 bracket over the drilled pilot holes to verify the 17 mm hole spacing matches.
 - ☐ Keep the 4 small wood screws in your pocket.
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4. Stop 3: College ECE Hardware Lab (Breadboard Requisition)

- **Location:** Basic Electronics / Microcontroller Lab.
- **Whom to Meet:** Lab Assistant / Lab in-charge.
- **Mission:** Borrow 1 standard **MB102 830-point solderless breadboard** (saves ₹80).

What to Say:

"Sir / Chetta, for our PBCST504 Microcontroller semester project hardware setup, could you please issue one standard 830-point breadboard from the lab inventory drawer? We will return it safely after our semester final evaluation."

- **Outcome A:** If they issue the breadboard, you are 100% done on campus!
 - **Outcome B:** If lab inventory is locked or unavailable, simply buy it from ET Store in Stop 4 for ₹80.
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5. Stop 4: Local Hardware Store OR ET Store Thrissur

- **Purpose:** Get the unbalance weight bolt, washers, and locknut.
- **Where to Go:** Any local retail nut-and-bolt hardware shop (Cheruthuruthy / Shoranur)
OR Emerging Technologies (ET Store, Ceeyel Tower near Thrissur Railway Station).

Exact Shopping List & What to Ask the Shopkeeper:

"Chetta / Bhai, I need small metric machine hardware:

- 1. One **M3 size, 16 mm or 20 mm bolt** (Star head or Allen key).*
- 2. Six **M3 flat steel washers** (for weight).*
- 3. Two **M3 Nyloc nuts** (the locknut with the white nylon plastic ring inside).*
- 4. (If breadboard was not obtained at college): One **MB102 830-point breadboard**."*

If Ordering via WhatsApp from ET Store (+91 9895241319):

Send them this message:

Hello ET Store, I need to pick up the following items from your Thrissur counter:

1. MB102 830-Point Breadboard (ET8102) - 1 pc (₹80.00) [Skip if got from college]
 2. M3 Lock Nut SS Sunloc (ET6602) - 4 pcs (₹8.24)
 3. M3 SS Washers (ET6159) - 10 pcs (₹6.00)
 4. M3x16mm Pan Head SS Bolt (ET6134) - 2 pcs (₹2.76)
 5. M3x20mm Pan Head SS Bolt (ET6107) - 2 pcs (₹2.90)
- Total: ~₹105. Please let me know if ready for pickup. Thank you!

6. What to Say if Faculty Ask Technical Questions (Cheat Sheet)

If a professor or lab instructor asks you about the setup, answer with these exact engineering reasons:

- **"Why do you need 80%–100% infill for a 3D-printed arm?"**

Answer:* "The arm carries an eccentric mass spinning at 600 RPM (10 Hz). Standard 15% infill would flex dynamically under centrifugal load, damping the vibration and distorting our FFT frequency harmonics."

- **"Why 10 mm to 12 mm thick plywood instead of acrylic or metal?"**

Answer:* "Plywood has high internal damping that suppresses high-frequency acoustic noise and tabletop rattle, while transmitting the low-frequency 10 Hz motor vibration cleanly into the ADXL345 sensor."

- **"Why use a Nyloc nut instead of a standard nut?"**

Answer:* "Under continuous 10 Hz vibration, a standard plain nut unthreads and flies off within 30 seconds. The internal nylon ring locks the threads securely without threadlocker glue."

- **"Why is the motor powered from a separate 12V adapter instead of the ESP32?"**

Answer:* "Galvanic isolation. The motor induces inductive kickback and brush noise that would crash the ESP32 or trigger brownout resets if grounds were shared."

7. Master File Directory for Amith's Phone / Pen Drive

All files are neatly arranged on the PC in your Downloads folder:

- **All-in-One WhatsApp ZIP:** [VibeGuard Complete Shareable Pack.zip](#)
- **FabLab Folder:**
[Downloads/VibeGuard Shareable Packs/01 College FabLab 3D Printing/](#)
- **Mech Lab Folder:**
[Downloads/VibeGuard Shareable Packs/02 College Mech Workshop Baseplate/](#)
- **Team Manual Folder:**
[Downloads/VibeGuard Shareable Packs/03 Team Assembly and Sourcing Manual/](#)